MSBA 511: Data Mining for Business Analytics

Conceptual Homework 3: Decision Trees

Objective

Be able to:

- Apply the steps of the decision tree algorithm to identify the best attribute for splitting the data, based on the Gini Index. Perform all Gini Index calculations by hand, and use these results to construct the decision tree for two levels.

Overview

A bank is evaluating credit card applications and using a decision tree for preliminary card approvals. The bank has collected a small dataset with three **predictor** variables: Employment Status (Employed/Unemployed), Credit Rating (Good/Bad), and Owns Property (Yes/No) and the **outcome** variable is Card Approved (Yes/No). The dataset is as follows:

Applicant	Employment Status	Credit Rating	Owns Property	Card Approved
1	Employed	Good	Yes	Yes
2	Employed	Good	No	Yes
3	Employed	Bad	Yes	Yes
4	Employed	Bad	No	No
5	Unemployed	Good	Yes	Yes
6	Unemployed	Good	No	No
7	Unemployed	Bad	Yes	No
8	Unemployed	Bad	No	No
9	Unemployed	Good	No	Yes
10	Employed	Good	No	Yes
11	Employed	Good	Yes	Yes
12	Unemployed	Good	No	Yes

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Tasks

1. Determine which attribute should be used for the **first split**. Perform all calculations using the Gini index. Clearly explain each step of your process and show all relevant computations.
2. Determine which attribute should be used for the **second split** to build a classification decision tree that predicts whether the card will be approved. Perform all calculations using the Gini index. Clearly explain each step of your process and show all relevant computations.
3. Build the decision tree with a max depth of 2 to build a classification decision tree that predicts whether the card will be approved.
    a. How does your model classify the following customer: Employment Status = Unemployed and Credit Rating = Good, and Owns Property = No and in **what confidence**?

Submission Requirements

Submit a PDF document to Gradescope that includes the following:

1. A table showing each split, variable, gini left/right values and combined impurity. **Note**: The split condition left (True) is when the variable = Unemployed, Bad, No (i.e. the non-positive condition for the variable).

Split	Variable	Gini Left (True)	Gini Right (False)	Combined Impurity
First				
Second				

2. Describe each step, and show all relevant calculations and include a screenshot or image of the decision tree.
    a. **Tip:** You are welcome to hand draw the decision tree or use the free https://app.diagrams.net/ to create the diagram.
3. Discuss **one** potential limitation of using this decision tree to make credit card approval decisions in this scenario